

Karthik Rao

| *Senior Product Leader - Production-scale AI, Data & Analytics*

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SKILLS

Product Management | Product Strategy | Vision & Roadmap | GTM | Prioritization | Competitive Analysis | Customer Insights | Product Development Lifecycle

Program & Project Management | Stakeholder Management | Cross-functional Leadership (Eng, DS, Design, Business) | Budgeting | Risk Management | Process Improvement

Data & Analytics | SQL | Python - Pandas | Power BI | Tableau, Excel | A/B Testing | EDA | ETL | RFM Analysis | Probability & Statistics | Metrics Design (OKR, KPI)

Machine Learning & AI | Regression | Classification | Logistics | Clustering | Tree based models | Time Series | Recommendation Ranking | NLP | Computer Vision | LLMs | RAG | Agents | Vector DB | Gen AI | AI governance

MLOps and /Tools: MLFlow | Docker | Cloud AWS | REST APIs | CI/CD | System Design | GitHub | Agile/Scrum | JIRA, Confluence | SDLC

EXPERIENCE

Independent Freelance Consultant, Self Employed

Nov 2020 - Present

- Advised FinTech clients on AI-driven analytics and GTM strategy, delivering scalable market intelligence dashboards for 100+ users and improving investment decision workflows through real-time operational insights.

Product Consultant, Sterna Security Devices Pvt Ltd, India

Feb 2024 - Mar 2025

Owned cloud-connected IoT security platform scaling from 0 to 10K+ users across 5 countries, reducing operational costs by 40% through platform automation and remote device management.

Tech: AWS IoT Core, Python, SQL, REST APIs, Docker, CI/CD

- Launched IoT security platform supporting 10K+ deployed devices, driving \$2M+ ARR within Year 1 through international market expansion
- Led GTM strategy across 5 countries, identifying 3 new market segments and driving 200+ qualified leads with 10% QoQ sales uplift
- Improved customer satisfaction from 4.0 to 4.4/5 through UX optimization and A/B testing, reducing support tickets by 25%
- Drove ISO 27001, GDPR, and CE compliance initiatives, reducing time-to-market by 30% and unlocking \$500K+ EU revenue opportunity

Digital Product Manager, Buckman Inc. USA

Jul 2020 - Oct 2020

Owned end-to-end IoT device management platform transforming fragmented legacy systems, scaling from 500 to 5K+ connected devices across 15 chemical manufacturing sites while reducing operational costs by 35%.

Tech: Azure IoT Core, Python, SQL, MQTT, REST APIs, Real-time Analytics, Docker

- Launched integrated IoT platform enabling seamless device onboarding and lifecycle management, supporting 5K+ smart meters and reducing provisioning time from 2 weeks to 2 days (85% faster)
- Secured executive approval for \$2M IoT modernization initiative by defining 18-month product roadmap with ROI-driven platform consolidation and scalability strategy.
- Built real-time IoT monitoring dashboards improving operational visibility and cross-team coordination, increasing CSAT from 3.5 to 4.4/5 and accelerating issue resolution workflows.
- Led smart meter rollout for hazard detection and asset monitoring across 15 manufacturing sites, preventing \$1M+ in potential safety incidents and reducing equipment downtime by 40%

Technical Product Manager, Dell Inc. USA

Dec 2015 - Nov 2019

Built enterprise KPI analytics platform from 0 to 500+ users across 10 departments, reducing reporting time by 70% and accelerating decision-making through unified real-time data infrastructure.

Tech: SQL, Python, REST APIs, Power BI, Klipfolio, Power Platform, Jenkins, Office 365

- Launched KPI analytics platform enabling real-time business metrics tracking for 500+ users, reducing manual reporting time from 20 hours/week to 6 hours/week (70% reduction) and saving \$500K annually in operational costs
- Secured C-suite approval for \$1.5M analytics modernization initiative by defining product strategy and ROI-driven 12-month roadmap, scaling adoption across 10 departments within 6 months.
- Led cross-functional team of 8 engineers and data analysts to build scalable REST API architecture, achieving 99.5% uptime and processing 10M+ data points daily
- Established KPI Center of Excellence standardizing enterprise metrics and automated dashboards, improving engineering delivery velocity by 15% and enabling data literacy training for 200+ employees.

Software Engineer, Dell Inc.

Dec 2008 - Nov 2015

Built enterprise test automation platform adopted across 50+ engineering teams, reducing release cycles from 4 weeks → 3 days (95% faster), cutting production defects by 40%, and saving \$800K annually in QA costs.

- Launched continuous testing platform reducing manual QA effort by 90% (200 hrs/sprint to 20 hrs/sprint) through scalable automated regression workflows.
- Drove organization-wide adoption achieving 80% automated test coverage within 12 months, enabling faster releases and reducing time-to-production by 60%

PROJECTS

Real-Time Retail Recommendation System (MVP) | Repo: github.com/Karthik-pmlead

- Built real-time recommendation system (Amazon/Netflix-style) using event-driven behavioral signals and hybrid ranking (ALS + session-aware collaborative filtering), improving simulated engagement (CTR/session duration) by 25% with sub-second WebSocket updates
- Designed real-time end-to-end AI experimentation pipeline with A/B testing and explainable recommendations, improving ranking quality (NDCG@10) by 20% through interpretable personalization models

AI Decision Intelligence Platform (MVP) | Repo: github.com/Karthik-pmlead

Decision intelligence system converting multi-source business signals into real-time operational insights and explainable recommendations.

- Built AI-powered decision intelligence platform processing 100K+ operational, customer, and business signals in real time, reducing manual analysis time by 60% and accelerating decision turnaround by 40% through orchestrated multi-engine analytics workflows
- Designed real-time event-driven AI orchestration system combining anomaly detection, sentiment analysis, trend analytics, and recommendation agents with live streaming updates and explainable reasoning pipelines

Operational Intelligence Platform (MVP) | Repo: github.com/Karthik-pmlead

- Built real-time multilingual sentiment intelligence platform processing 100K+ review events, achieving 94% sentiment classification accuracy with <300ms latency for e-commerce feedback analytics.
- Designed AI analytics system with streaming dashboards and MLOps observability, improving customer response time by 50% through actionable sentiment trends, topic insights, and real-time alerting.

EDUCATION

- **MBA, Management of Technology**, Georgia Institute of Technology, USA, 2013
- **M.S Electrical Engineering** (Major: Computer Science) Wichita State University, USA, 2008
- **B.E Electrical Engineering**, Anna University, India, 2005

CERTIFICATIONS

- **Scaler** - Data Science and Machine Learning, 2026
- **Project Management** Course, 2025
- **Product School** - Product Management Certificate, 2020
- **Udacity** - Data Analyst, Data Engineering, AI Product Manager, AWS Cloud Architect, 2020-22